

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

IV Semester of (BSc) Microbiology Examination April/ May 2018

Course Code & Name MI 401 & Microbial Genetics

Date: 9/5/2018 Day: Wednesday Time: 10:00 am To 10:20 am Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	Choose the correct answer for the following questions.		10
1.	_____ codes for monocystronic m-RNA.		01
	(i) <i>E. coli</i>	(ii) <i>Bacillus</i> sp.	
	(iii) <i>S. cerevisiae</i>	(iv) Both, (i) and (ii)	
2.	The <i>trp</i> operon gets switched on/ off by _____.		01
	(i) <i>Trp</i> repressor	(ii) Tryptophan concentration	
	(iii) Attenuation	(iv) All of the above	
3.	In _____ mutation, the change in the phenotype would not be seen.		01
	(i) Silent mutation	(ii) Lethal mutation	
	(iii) Neutral mutation	(iv) Both, (i) and (iii)	
4.	If Adenine gets replaced by Guanine due to mutation, it is called as _____		01
	(i) Transition	(ii) Transversion	
	(iii) Transformation	(iv) None of the above	

5.	_____ is a chemical way to transfect DNA.		01
	(i) Use of dendrimers	(ii) Cell squeezing	
	(iii) Liposomes	(iv) Both, (i) and (iii)	
6.	<i>LacZ</i> codes for _____		01
	(i) Permease	(ii) Transacetylase	
	(iii) β -galactosidase	(iv) All of the above	
7.	_____ is the longest transposable element known till date.		01
	(i) IS elements	(ii) Composite transposons	
	(iii) Non-Composite transposons	(iv) Bacteriophage Mu	
8.	Following is incorrect about plasmids _____		01
	(i) Plasmids are self-replicating in nature.		
	(ii) Plasmids are extrachromosomal DNA.		
	(iii) All plasmids contain 'cos site'		
	(iv) None of the above		
9.	When the phage DNA gets inserted within the bacterial genome, it is called as ____.		01
	(i) Endophage	(ii) Prophage	
	(iii) Interphage	(iv) None of the above	
10.	_____ can be used to study fungal linkage map.		01
	(i) Ordered tetrads	(ii) Unordered tetrads	
	(iii) Both, (i) and (ii)	(iv) None of the above	

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Course Code & Name MI 401 & Microbial Genetics

Date: 9/5/2018 Day: Wednesday Time: 10:20 am To 12:00 Noon Maximum Marks: 25

Instructions:

1. Section I and II must be attempted in SINGLE ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION –I		
Q - II Answer the following questions as directed. (Any 5)		10
1	What is the Central Dogma of Life	02
2	Define: tetrads	02
3	Differentiate: Ordered tetrads and unordered tetrads	02
4	Define: Transfection. Enlist any five methods of transfection	02
5	Enlist various types of mutations.	02
6	Differentiate: conservative, replicative and retrotransposons	02
SECTION – II		
Q-III Answer the following questions as directed (Any 3)		15
1	What would be the impact on the lac operon under following conditions? I. Lactose is absent II. Lactose is present III. Glucose and Lactose, both are present	05
2	Short note: <i>Vir</i> genes and their role in Ti-plasmid infection	05
3	Elucidate: Generalized and Specialized Transduction	05
4	Define: Conjugation. Explain mating between... I. $F^+ \times F^-$ II. $Hfr \times F^-$ III. $F' \times F^-$	05